

Lac-Phe (N-lactoyl phenylalanine): study discovery, exercise effect, and MoTrPAC rat alignment

Live-mode report. Every row below comes from a public released record fetched through the declared network boundary; no synthetic stand-ins are used, and unavailable data is reported as a coverage gap.

Headline

- Human exercise effect is present and large. In MW ST003662 (healthy male and female athletes, whole blood, post- vs pre-exercise), Lac-Phe rises log2FC +1.16 (95% CI +0.91 to +1.41; ~2.2-fold), paired t p=2.73e-15, BH FDR=2.50e-14, n=116 pairs.
- Rat MoTrPAC alignment is blocked by feature coverage, not by effect size. Lac-Phe does not appear anywhere in pass1b-06 metabolomics: 0 matches across 2459 unique features, 99,236 timewise rows, 19 tissues, 4 training-week groups.
- No N-lactoyl amino-acid conjugate of any kind is in the rat feature space, so this is a panel-coverage gap for the whole conjugate class rather than a missing single annotation.
- The effect is individually consistent, not a group artefact. Lac-Phe increases in 92 of 116 paired athletes (79%), and it is the 4th largest of 165 measured effects - behind only Lactic acid, Pyruvic acid, Xanthine.
- Lactate rise explains only part of it. Within-participant delta log2 lactate vs delta log2 Lac-Phe gives $r = 0.35$ ($p = 1.9e-03$, $n = 75$), so the conjugate is not a simple readout of substrate availability.
- The single-study limit is a deposition limit, not a literature limit. A literature sweep of Europe PMC, Crossref and bioRxiv/medRxiv returned 375 deduplicated records, 100 of which name Lac-Phe in the retrieved title or abstract; 19 are human exercise records and 5 of those describe an interventional or randomized design. None names a repository accession in its abstract, so their deposition status is unresolved rather than absent (Aim 4).
- Alignment therefore stops at precursor-level evidence (lactic acid, phenylalanine) plus a documented availability gap. It is not a cross-species replication and must not be reported as one.

Chemical identity of the query

field	value
query submitted to MW	N-Lactoyl phenylalanine
name variants also searched	Lac-Phe, N-lactoylphenylalanine
resolved RefMet name	N-Lactoyl phenylalanine
RefMet ID	RM0131640
formula / exact mass	C12H15NO4 / 237.100109
InChIKey	IIRJJZHHNGABMQ-WPRPVWTQSA-N
PubChem CID	11075454
RefMet class path	Organic acids > Amino acids and peptides > Amino acids
resolution source	https://www.metabolomicsworkbench.org/rest/refmet/name/N-Lactoyl%20phenylalanine/all

Identity is resolved at the RefMet-name and structure-annotation level. It is not MSI level-1 confirmation in any of the assays below; every effect row stays requires_human_review for assay identity.

Aim 1 - Study discovery

Metabolomics Workbench metstat was queried for the resolved RefMet name across all species, sources and platforms. 57 studies / 60 analyses report Lac-Phe.

species	analyses
-----	-----
Human	24
Mouse	20
Cow	2
Rat	2
not_reported	1
Akkermansia muciniphila	1
Escherichia coli	1
Gardnerella vaginalis	1
Gemella morbillorum	1
Horse; Yak; Donkey; Camel	1
Lactobacillus rhamnosus	1
Macaque monkey	1
Pig	1
Sheep	1
Staphylococcus aureus	1
Wheat	1

- Human blood analyses: 11 across 9 studies.
- Of those, exactly one is an exercise-physiology design: ST003662. Six of the remaining eight studies are disease cohorts (cancer x3, ARDS, ALS, multiple sclerosis); the other two are a clinical-prediction modelling study (ST003587) and a healthy-volunteer plasma/faeces study (ST004826). None of the eight carries an exercise exposure, so none can answer an exercise question.
- Rat MW analyses exist (2) but are oxycodone-exposure plasma and post-colectomy feces designs - neither is an exercise design.
- Discovery verdict: the exercise-relevant human Lac-Phe evidence base in MW is a single study. Treat every conclusion below as single-study *deposited* evidence. Aim 4 shows the published literature is not single-study, so the limit here is MW deposition and discoverability rather than measurement.

Full table: data/live/metabolite_search_lacphe/metstat_metabolite_study_hits.csv

Aim 2 - Human exercise effect (MW ST003662)

- Study: Metabolome trajectory of exercise physiology- a comprehensive study of healthy male and female athletes
- Species: Homo sapiens; license CC BY 4.0
- Matrix: whole blood, umol/L (analyses AN006015 HILIC + AN006016 reversed phase)
- Contrast: post-exercise vs pre-exercise (Collectionpoint After vs Before); orientation: positive log2fc = higher post-exercise
- Statistic: paired t-test on log2 abundance, BH-adjusted within stratum
- Named metabolites in the fetched panel: 178
- Features with a usable paired estimate (>=3 complete pairs): 165

stratum	log2FC	95% CI	fold-change	p	BH FDR	pairs	mean pre	mean post
-----	-----	-----	-----	-----	-----	-----	-----	-----
All athletes	+1.16	+0.91 to +1.41	2.23x	2.73e-15	2.50e-14	116	0.539	0.755
Male	+1.37	+1.01 to +1.73	2.59x	2.49e-10	2.41e-09	63	0.562	0.764
Female	+0.91	+0.57 to +1.25	1.87x	2.07e-06	1.53e-05	53	0.512	0.744

Reading: Lac-Phe is one of the strongest post-exercise increases in this blood metabolome, and it is significant in both sexes. The male point estimate is larger than

the female one, but the sex strata are nested inside the all-athlete stratum, share the platform and batch, and their CIs overlap - this is not evidence of a sex-by-exercise interaction. A formal interaction test on the sample-level matrix is required before any sex-difference claim.

- Figure: figures/human_ST003662_lacphe_volcano.png
- Figure: figures/human_ST003662_lacphe_effect_by_sex.png
- Table:
data/live/metabolite_scan/n_lactoyl_phenylalanine/mw_queried_metabolite_effects.csv

Aim 3 - MoTrPAC rat alignment

What was searched

- Endpoint: https://search.motrpac-data.org/api/search_public, query {"study": "pass1b06", "omics": "metabolomics", "size": 200000}
- Block: metabolomics_timewise (trained vs sedentary control, timewise differential rows)
- Rows returned: 99,236; unique features: 2459
- Tissues: adrenal, brown adipose, colon, cortex, gastrocnemius, heart, hippocampus, hypothalamus, kidney, liver, lung, ovaries, plasma, small intestine, spleen, testes, vastus lateralis, vena cava, white adipose
- Training-week groups: 1w, 2w, 4w, 8w
- Name columns scanned: refmet_name, metabolite, feature_id

Result: coverage gap

0 rows matched Lac-Phe. Name matching was normalization-insensitive (punctuation and spacing stripped, so N-Lactoyl phenylalanine, N-lactoylphenylalanine and lactoylphenylalanine all collapse to one key). A direct substring sweep of all three name columns (refmet_name 2459 unique, metabolite 2651 unique, feature_id 2739 unique) returns zero rows containing lactoyl - no N-lactoyl-phenylalanine, -leucine, -valine, or any other conjugate.

This is an availability gap in the rat metabolomics panel, and it is the correct scientific finding to report. It is emphatically *not*: (a) evidence that Lac-Phe is unchanged by training in rats, (b) grounds for substituting a synthetic or imputed rat Lac-Phe row, or (c) a reason to swap in a nearby analyte and call it Lac-Phe.

Closest available rat evidence - precursors only

The two substrates of the Lac-Phe conjugation reaction *are* measured in rat:

metabolite	tissues in focus set	timewise rows	rows adj_p<0.05	rows nominal p<0.05
lactic acid	gastrocnemius, heart, liver, plasma	32	0	9
phenylalanine	gastrocnemius, heart, liver, plasma, vastus lateralis	40	0	3

No precursor cell in the five focus tissues reaches adj_p<0.05. Across all 19 tissues the picture is not quite empty, and the report states it explicitly rather than rounding it to zero:

- 1 of 214 precursor cells survives FDR: phenylalanine in male white adipose at 8w

(logFC +0.37, adj_p 0.018).

- 23 cells are nominally significant without surviving FDR. For 4 of those the source reports no adjusted p at all (heart), so they cannot be assessed against FDR either way; the 19-tissue heatmap marks them with a grey dagger.
- That single surviving signal is phenylalanine in white adipose, not muscle or plasma, and it is a free-amino-acid training effect. It is precursor evidence about substrate availability in that tissue; it is not evidence about Lac-Phe, which is absent from the panel entirely.
- Figure: figures/rat_pass1b06_lacphe_precursors.png (five focus tissues), figures/f6_rat_precursor_tissue_heatmap.png (all 19 tissues, with the FDR-significant cell marked)
- Precursor abundance changes constrain substrate availability, not conjugate formation. CNBP2-mediated Lac-Phe synthesis is a separate step, and neither substrate is a validated proxy for the conjugate.

The gap is class-wide, not a single missing annotation

All 22 N-lactoyl amino-acid conjugates in RefMet were searched in MW and checked against each exercise source's feature space. 17 of 22 are reported somewhere in MW, and Lac-Phe is the best covered of them (60 analyses).

conjugate	MW analyses	in ST003662 human whole blood	in MoTrPAC human public plasma	in MoTrPAC rat pass1b-06
N-Lactoyl phenylalanine	60	yes	no	no
N-Lactoyl leucine	31	no	no	no
N-Lactoyl tyrosine	28	no	no	no
N-Lactoyl methionine	28	no	no	no
N-Lactoyl valine	21	no	no	no
N-Lactoyl isoleucine	17	no	no	no
N-Lactoyl tryptophan	11	no	no	no
N-Lactoyl histidine	8	no	no	no
N-Lactoyl glutamic acid	1	no	no	no
N-Lactoyl glutamine	1	no	no	no
N-Lactoyl glycine	1	no	no	no
N-Lactoyl arginine	1	no	no	no
N-Lactoyl lysine	1	no	no	no
N-Lactoyl aspartic acid	1	no	no	no
N-Lactoyl serine	1	no	no	no
N-Lactoyl threonine	1	no	no	no
N-Lactoyl asparagine	1	no	no	no
N-Lactoyl DOPA	0	no	no	no
N-Lactoyl cysteine	0	no	no	no
N-Lactoyl citrulline	0	no	no	no
N-Lactoyl ornithine	0	no	no	no
N-Lactoyl alanine	0	no	no	no

Every conjugate is absent from both MoTrPAC feature spaces - the rat pass1b-06 panel (19 tissues) and the public human plasma acute panel (1657 features). Only Lac-Phe appears in ST003662. The implication is a panel-design gap for the whole conjugate class rather than a per-metabolite annotation miss, which is what a future data request would have to address.

- Figure: figures/f5_lactoyl_class_coverage.png

Barriers to alignment

barrier	human ST003662	rat pass1b-06	consequence
feature coverage	Lac-Phe measured	Lac-Phe absent	no shared analyte: crosswalk impossible
estimand	acute post- vs pre-bout, paired within	chronic 1-8w training vs sedentary, bet...	different biological question even if th...
matrix	whole blood, umol/L	plasma and 18 solid tissues, platform-s...	units and matrix not exchangeable
statistic	paired t on log2 umol/L, computed here	consortium timewise model logFC	effect scales not directly comparable

Even a future rat panel that adds Lac-Phe would still face the estimand and matrix barriers. Alignment would then require an effect-level synthesis with study-specific effects, not a pooled analysis.

- Figure: figures/lacphe_cross_species_alignment.png

Aim 4 - Published literature

What was searched

- Indexes: Europe PMC (which indexes MEDLINE/PubMed, PMC and preprints), PubMed E-utilities, Crossref, and the bioRxiv/medRxiv detail API for preprint version and journal linkage.
- Europe PMC expression: ("N-Lactoyl phenylalanine" OR "Lac-Phe" OR "N-lactoylphenylalanine" OR "lactoylphenylalanine" OR "N-lactoyl-phenylalanine")
- Records retrieved: 464 rows across all indexes, 375 after cross-source deduplication on DOI, PMID and PMCID.

index	status	reported hits	retrieved	complete sweep
-----	-----	-----	-----	-----
europe_pmc	ok	306	304	yes
pubmed	unavailable	unknown	0	no
crossref	ok	47,194	160	no
biorxiv_medrxiv	ok	6	6	yes

Retrieval limits that bound every count below

- pubmed could not be reached from this network egress. Its coverage for this query is unknown. This is an availability gap, not a finding that it holds no matching publication, and the Europe PMC SRC:MED subset is the only MEDLINE coverage this report actually has.
- crossref returned a ranked relevance sample out of a scored candidate pool of 47,194 summed across the five variant queries, so absence of a paper from the tables below is not evidence that the index does not hold it.
- Europe PMC index reported 306 hits but returned 304 records across all pages; the difference is unexplained by the index and the shortfall is not a screening decision.
- Bibliographic records only. No full text was read, so data availability statements, methods-level assay identity, and reported effect sizes are outside what this lane can establish.

A name match is not subject evidence

Of 375 deduplicated records, 100 actually name Lac-Phe or a declared variant in the retrieved title or abstract. 220 do not - they matched on full-text indexing or on Crossref relevance ranking - and 55 were returned with no abstract at all, so their subject could not be located either way. All of those are escalated as unconfirmed rather than counted as evidence.

Lac-Phe is also used for lactide-phenylalanine copolymers, so a string match in the materials literature denotes a different chemical entity. Among the subject-named records, 0 carry materials-science context with no biological context and 5 carry both. Both classes are escalated for entity confirmation from full text rather than counted as

subject evidence; the automated signal is deliberately conservative, so it under-detects rather than over-flags.

What the subject-confirmed literature contains

evidence class	records	note
human exercise	19	18 are peer-reviewed primary research
animal or in-vitro mechanistic	13	background only; never a human required-term match
human, non-exercise context	18	disease, drug, diet and assay-method reports
secondary synthesis	19	reviews and meta-analyses; not primary evidence
retracted or withdrawn	2	excluded from primary evidence and named below
unscreenable (all records, not only subject-confirmed)	142	56 returned no abstract and 86 name no species in the abstract; unresolved, no...

- Retraction-flagged: *The causal relationship between blood metabolites and rosacea: A Mendelian randomization* (Skin research and technology : official journal of Intern..., 2024). Verify the retraction status before any use.
- Retraction-flagged: *Withdrawn: Combinatorial lipidomics and proteomics underscore erythrocyte lipid membrane aberrations in the development of adverse cardio-cerebrovascular complications in maintenance hemodialysis patients* (Redox biology, 2024). Verify the retraction status before any use.

This qualifies the Aim 1 discovery verdict

Aim 1 found exactly one exercise-design human study reporting Lac-Phe in Metabolomics Workbench. The published record is larger: 19 subject-confirmed human exercise records, of which 5 report an interventional or randomized exercise design.

year	design	journal	study
2022	interventional exercise bout or program	Metabolites	Exercise-Induced N-Lactoylphenylalanine Predicts Adipose Tissue Loss d...
2025	randomized controlled trial	Metabolomics	Exercise intensity determines circulating levels of Lac-Phe and other ...
2025	interventional exercise bout or program	Biology of sport	N-Lactoyl amino acids as metabolic biomarkers differentiating low and ...
2026	randomized controlled trial	EMBO molecular medicine	The anti-obesogenic metabolite, Lac-Phe, is elevated by metformin trea...
2026	interventional exercise bout or program	BMC Endocrine Disorders	The effect of blood flow restriction training on abdominal visceral fa...

Counts here are records, not resolved studies: identifiers and preprint-to-journal linkage are merged, but companion papers, secondary analyses and cohort overlap between records are not resolved, so the number of distinct human exercise cohorts is at most this and may be fewer.

So *single-study* is a true statement about the MW-deposited evidence base and a false one about the literature. The bottleneck is deposition and discoverability, not measurement: independent human exercise studies have measured Lac-Phe, but their data are not retrievable through the MW metstat lane that Aim 1 searched.

That said, 0 of 19 subject-confirmed human exercise records name a repository accession anywhere in their retrieved bibliographic text. An abstract is not a data availability statement, so this does not establish that those studies deposited nothing - it establishes that deposition cannot be resolved without reading their full texts. The distinction is the difference between a confirmed deposition gap and an unread one, and this report claims only the latter.

Direction of effect is also not claimed from this lane. The retrieved abstracts state their own results, which is bibliographic testimony rather than a re-analysis; no effect size from any of these records was recomputed here, and none may be pooled with the ST003662 estimate above.

Escalations raised by the literature lane

419 escalations, by type:

escalation	records
-----	-----
subject name not located in record	220
unscreenable record	142
unresolved data deposition	30
cross source metadata conflict	9
preprint without peer review	9
possible name homonym	5
retraction flag in publication type	2
retrieval source unavailable	1
retrieval truncated	1

Full report: reports_live/literature/n_lactoyl_phenylalanine_literature_report.md. Full queue: data/live/literature/lacphe/literature_escalations.csv.

Figure suite

Rendered by scripts/render_lacphe_figures.py (offline) with one backing CSV per figure under reports_live/lacphe/figure_tables/.

figure	what it shows	the claim it supports
-----	-----	-----
figures/f1_lacphe_participant_response.png	every paired athlete's pre -> post Lac-Phe, by ...	the increase is individual and near-uniform (9...
figures/f2_lacphe_effect_ranking.png	all 165 features ranked by effect	Lac-Phe is rank 4; the features above it are i...
figures/f3_lacphe_lactate_coupling.png	delta lactate vs delta Lac-Phe within participants	substrate rise tracks conjugate rise (r = 0.35...
figures/f4_sex_concordance.png	male vs female effect for every feature	internal consistency (r = 0.89, 127/161 same d...
figures/f5_lactoyl_class_coverage.png	22 conjugates: MW analysis counts (all species...	the coverage gap is class-wide
figures/f6_rat_precursor_tissue_heatmap.png	rat precursor logFC across 19 tissues x 4 week...	substrates are measured (phenylalanine 19/19 t...
figures/f7_discovery_landscape.png	the 60 MW analyses by species and human sample...	Lac-Phe is widely reported but almost never un...
figures/human_ST003662_lacphe_volcano.png	whole-metabolome volcano per stratum	effect size and significance in context
figures/human_ST003662_lacphe_effect_by_sex.p...	forest of stratum effects with 95% CIs	overlapping CIs across sexes
figures/rat_pass1b06_lacphe_precursors.png	precursor logFC in the five focus tissues	precursor-level evidence only
figures/lacphe_cross_species_alignment.png	availability status per source	alignment is blocked by coverage, not effect s...

Figure plates

The same figures, inline, so the markdown and the PDF render carry the evidence rather than pointing to it. Each has a backing CSV of the exact plotted values in figure_tables/.

Lac-Phe rises in 92 of 116 athletes (79%) after a single exercise bout
MW ST003662 · N-Lactoyl phenylalanine (RefMet RM0131640) · paired whole-blood samples

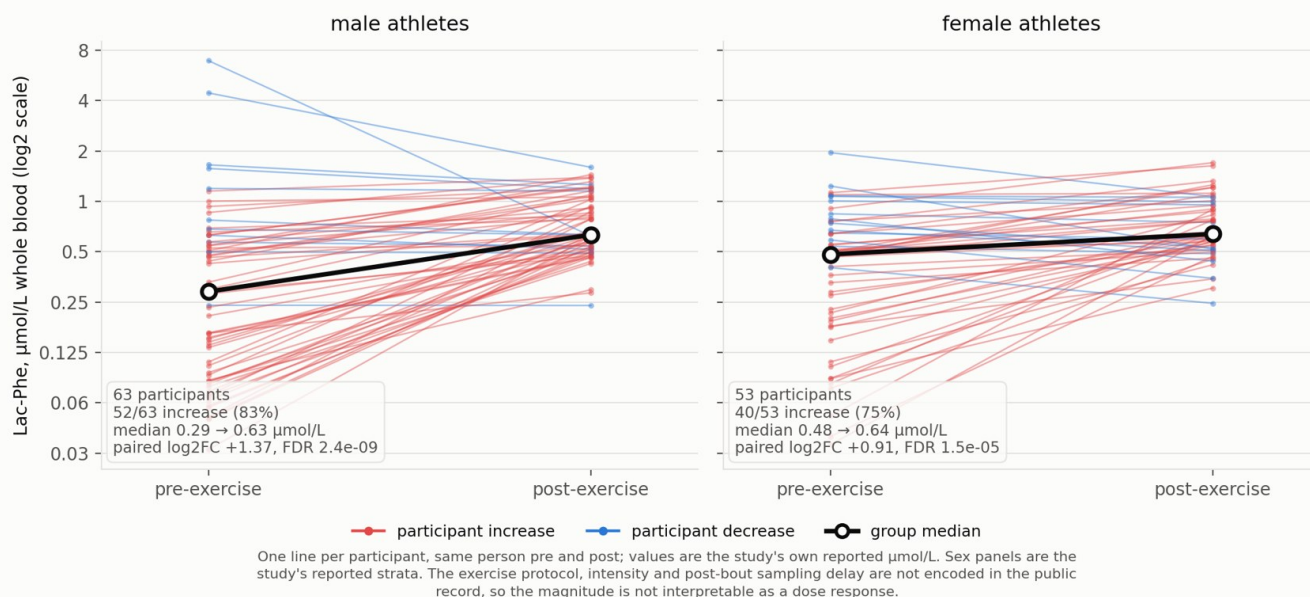


Figure 1. Lac-Phe pre- vs post-exercise in every paired athlete of MW ST003662, by sex. 92 of 116 participants increase.

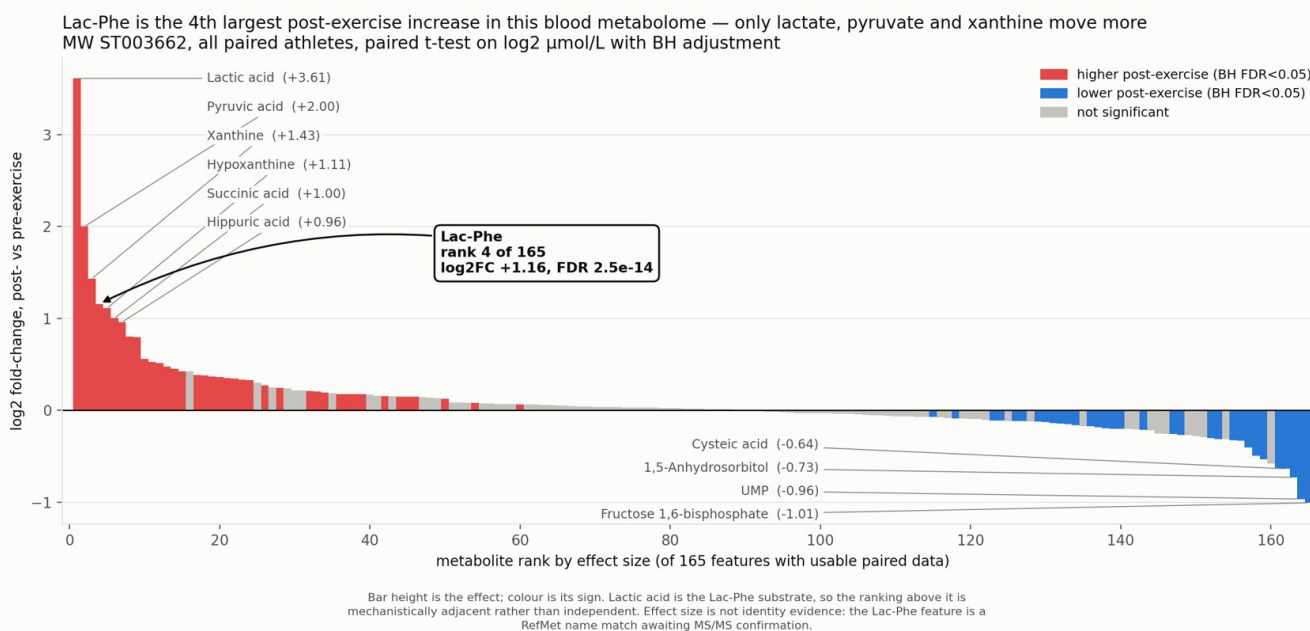
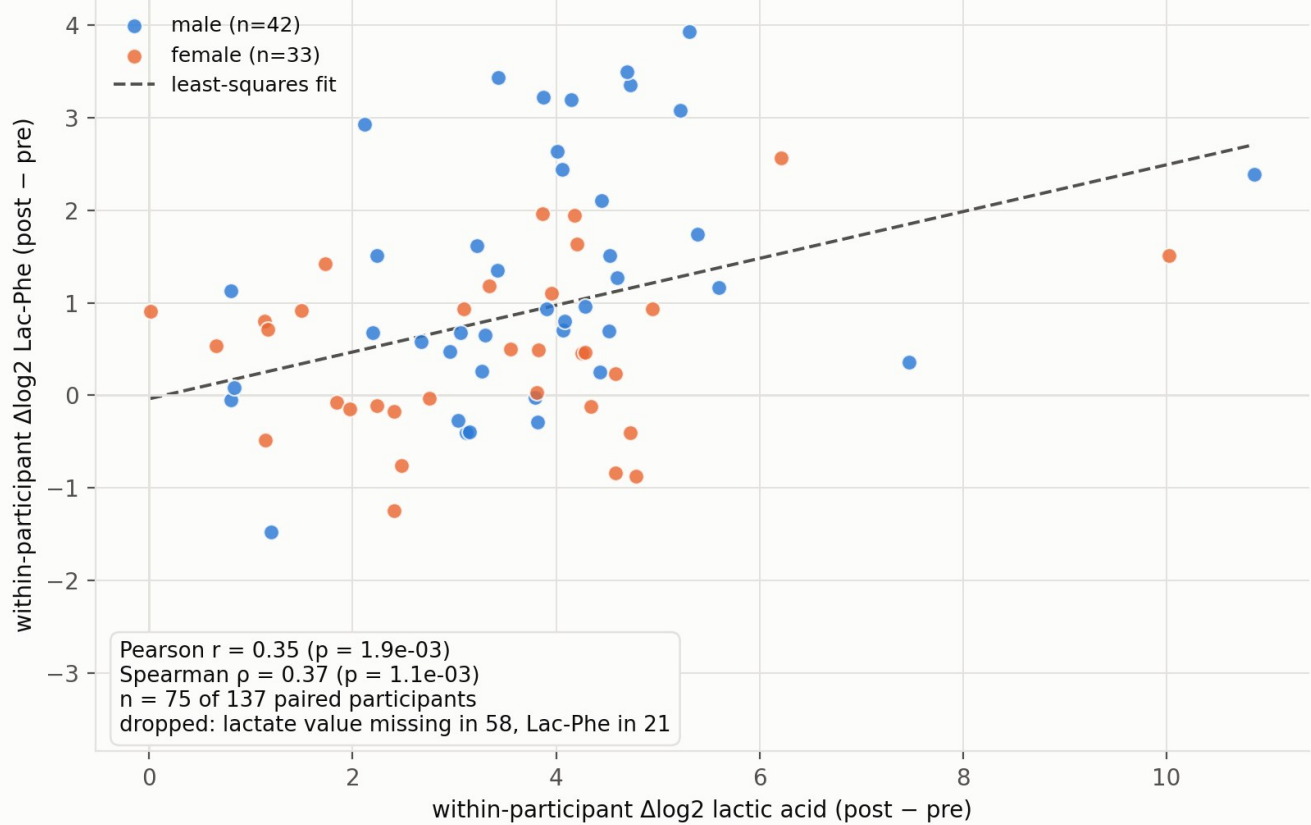


Figure 2. All 165 features ranked by post- vs pre-exercise effect. Lac-Phe is rank 4.

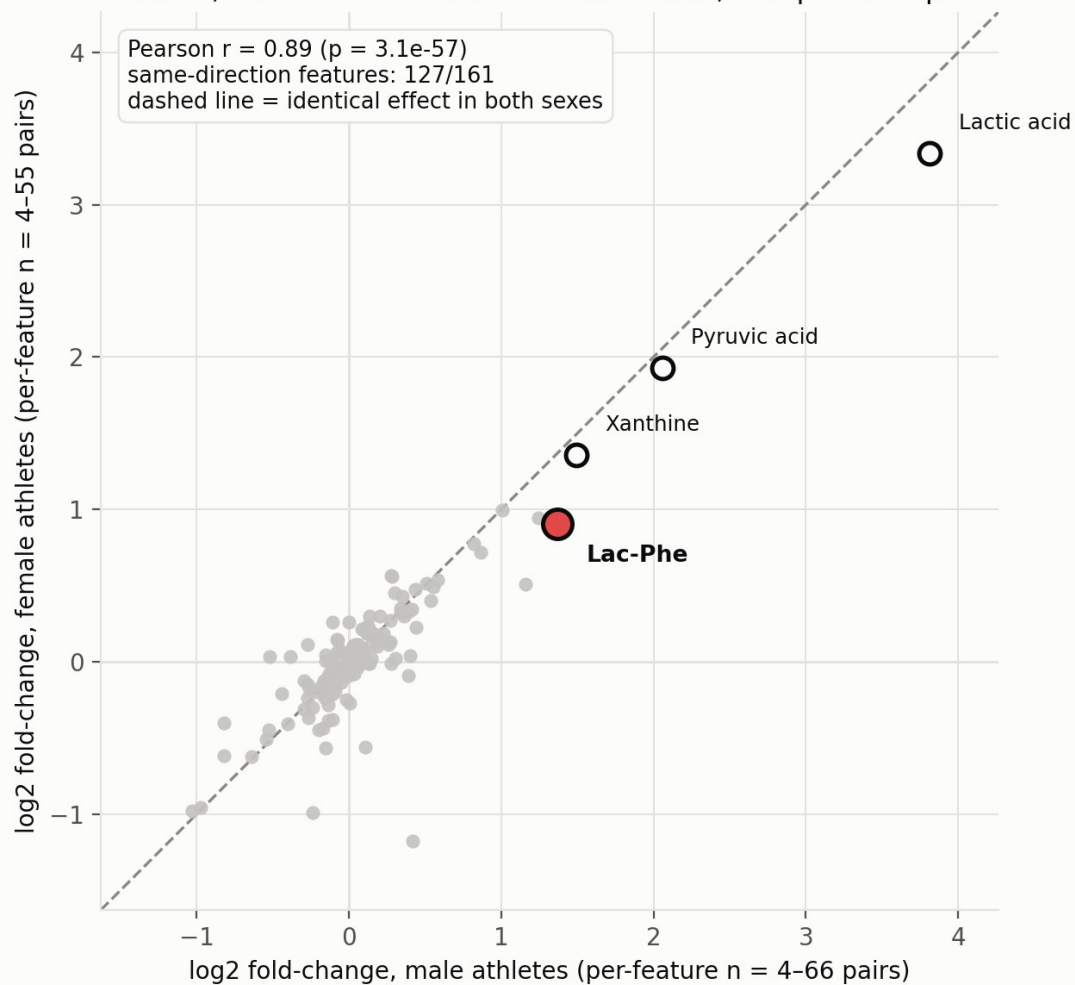
A larger lactate rise tracks a larger Lac-Phe rise, but explains only part of it ($r = 0.35$)
 MW ST003662 whole blood, same individuals on both axes



Association within one study, not a mechanism test: CNDP2-mediated conjugation was not measured here, and lactate is only one of the two substrates. Both axes are within-participant changes, so between-person baseline differences are removed by construction. The plotted subset is NOT a random sample: lactic acid has no reported pre-exercise value in most dropped participants, so a selection effect on this correlation cannot be excluded.

Figure 3. Within-participant lactate change vs Lac-Phe change ($r = 0.35$, $n = 75$ of 137; the dropped participants mostly lack a reported pre-exercise lactate value, so the subset is not random).

Male and female effect estimates agree across the measured metabolome
MW ST003662, 161 features tested in both strata, independent paired tests



Both axes come from the same study, platform and batch, so this is internal consistency, not independent replication. Pairs are dropped per feature on missingness, so the low- n tail (features estimated from as few as 3-4 pairs) carries more scatter than the bulk. Lac-Phe sits to the right of the identity line, i.e. a larger male point estimate, but the strata are not an interaction test — a sex difference needs a model on the sample-level matrix.

Figure 4. Male vs female effect estimates for the 161 features tested in both strata ($r = 0.89$). Same study and platform, so this is internal consistency, not independent replication.

The whole N-lactoyl amino-acid class is missing from both MoTrPAC exercise panels
22 RefMet conjugates · Lac-Phe is the best covered of them in MW, and still absent from MoTrPAC human and rat feature spaces

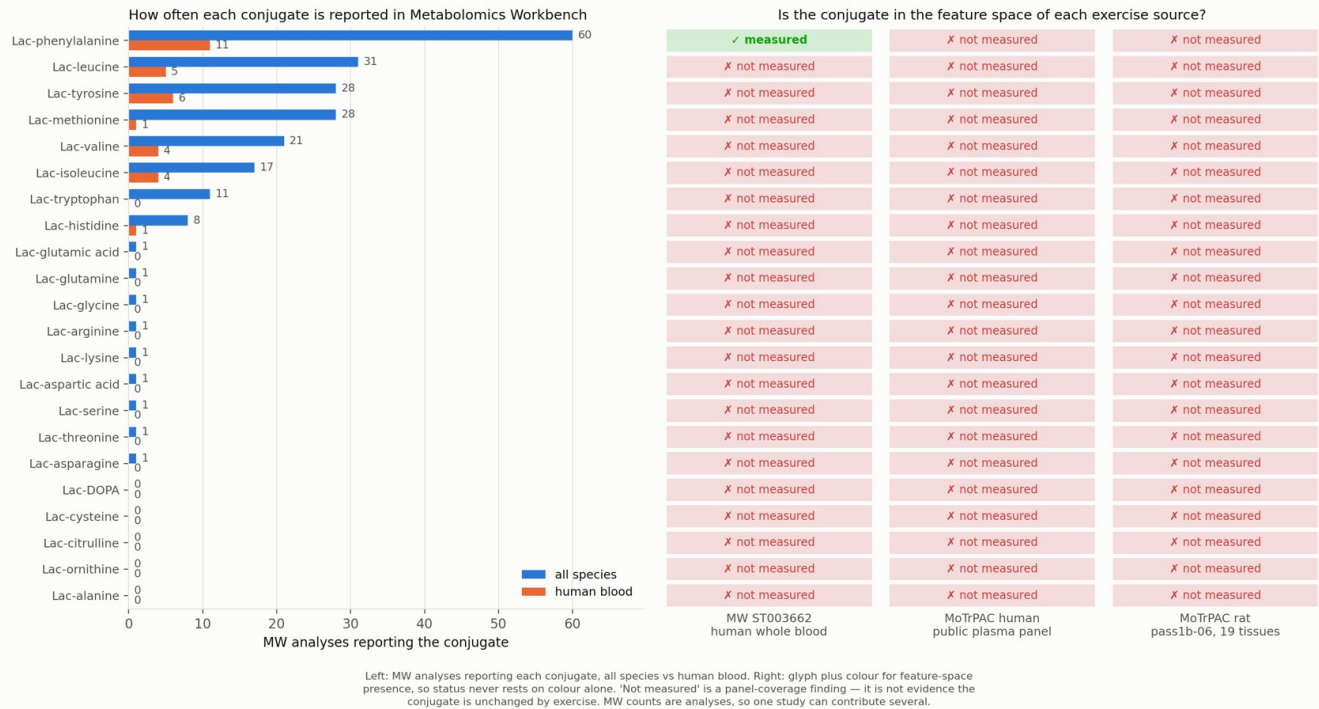


Figure 5. All 22 N-lactoyl amino-acid conjugates in RefMet: MW analysis counts, and presence in the three exercise feature spaces. The coverage gap is class-wide.

Rat pass1b-06 reports both Lac-Phe substrates — and the conjugate itself in no tissue at all
phenylalanine in 19/19 tissues, lactic acid in 9/19, any N-lactoyl conjugate in 0/19 (0 of 2,459 rat features)

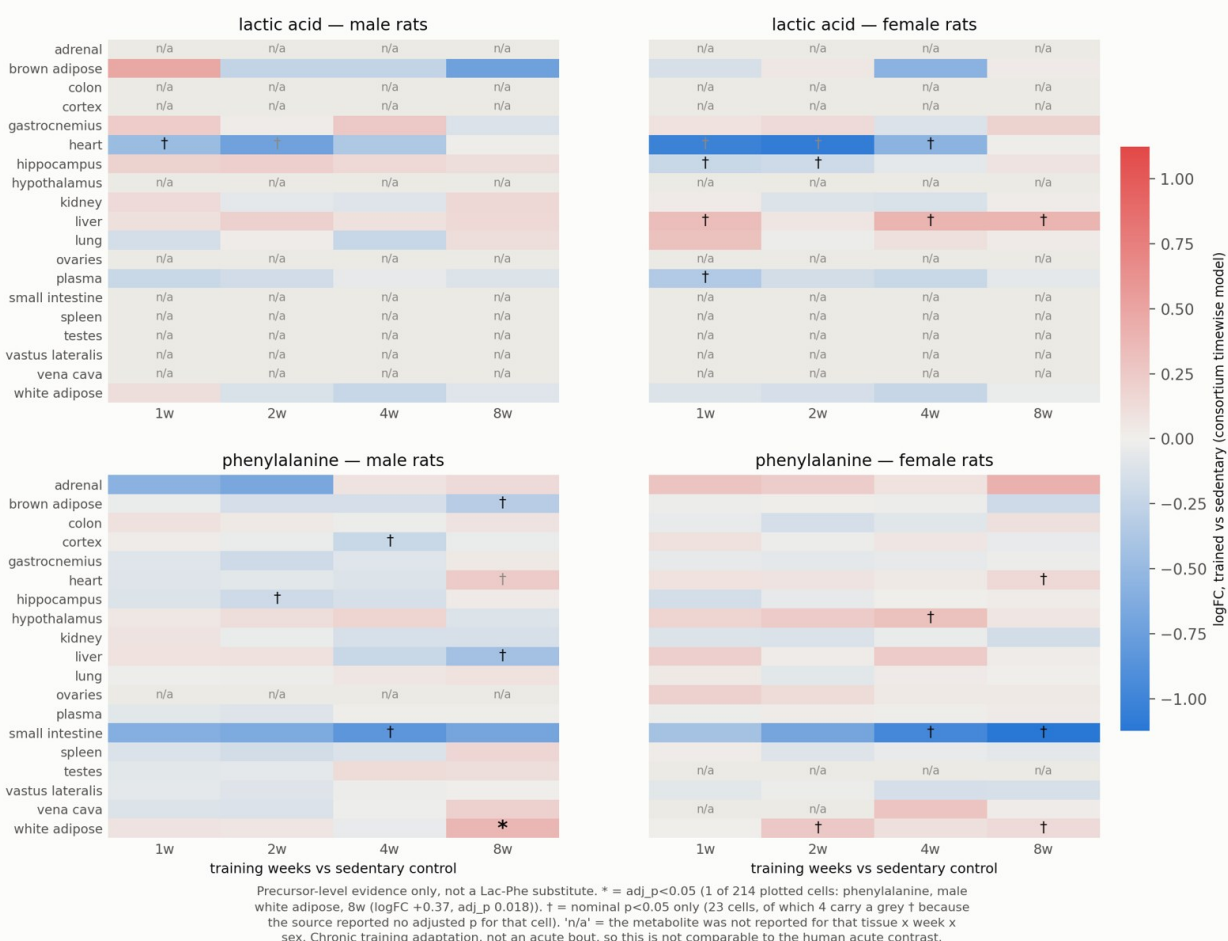


Figure 6. Rat pass1b-06 precursor logFC across 19 tissues, four training weeks and both sexes. One of 214 cells survives FDR (*); Lac-Phe itself is absent from every tissue.

Lac-Phe is widely reported but almost never in an exercise design

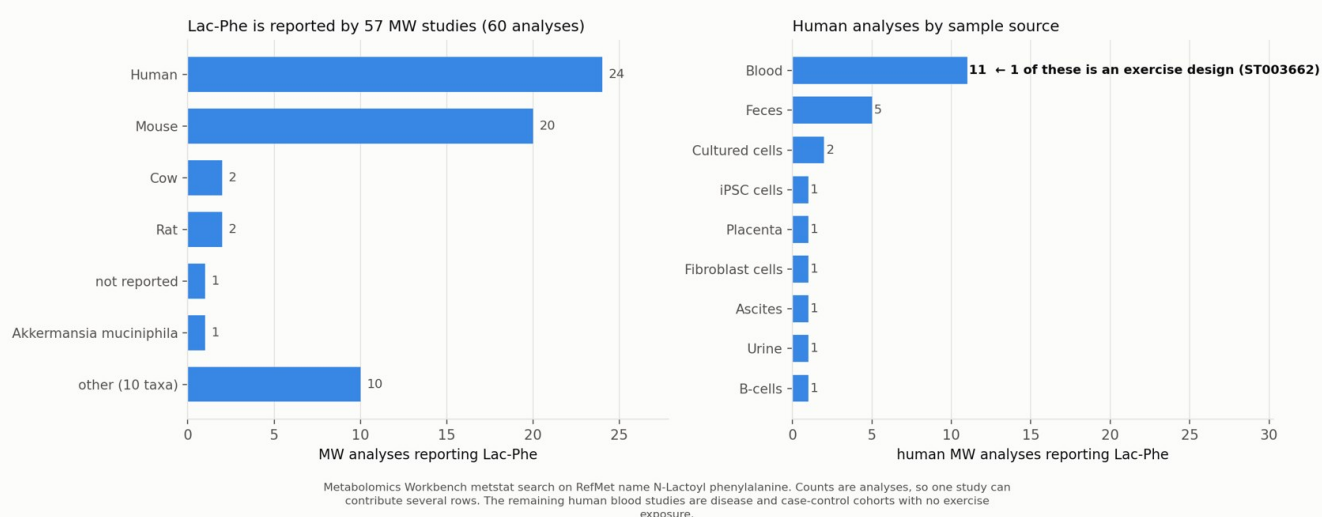


Figure 7. The 60 MW analyses reporting Lac-Phe, by species and by human sample source.

Human whole-blood metabolome, post- vs pre-exercise — MW ST003662 (athletes)
 Lac-Phe = N-Lactoyl phenylalanine (RefMet RM0131640); paired t-test on log₂ μmol/L, BH within stratum

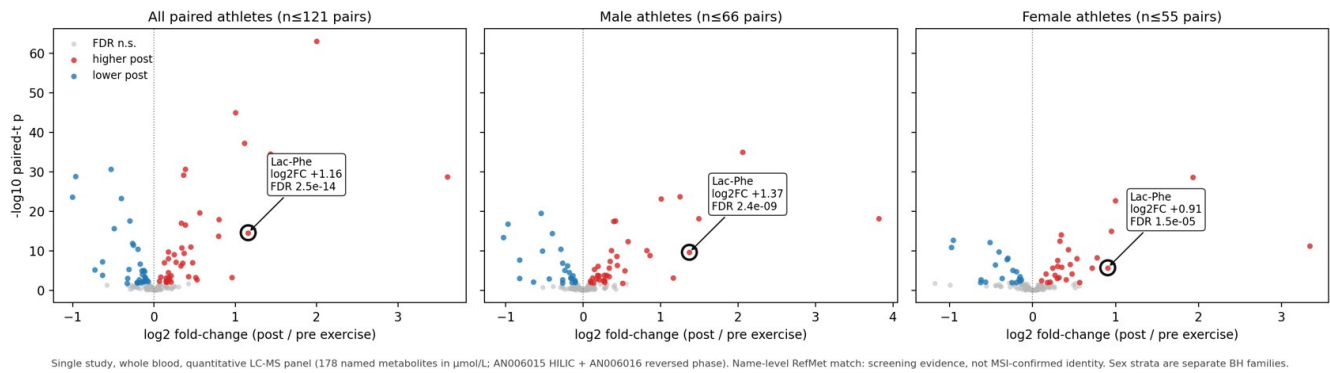
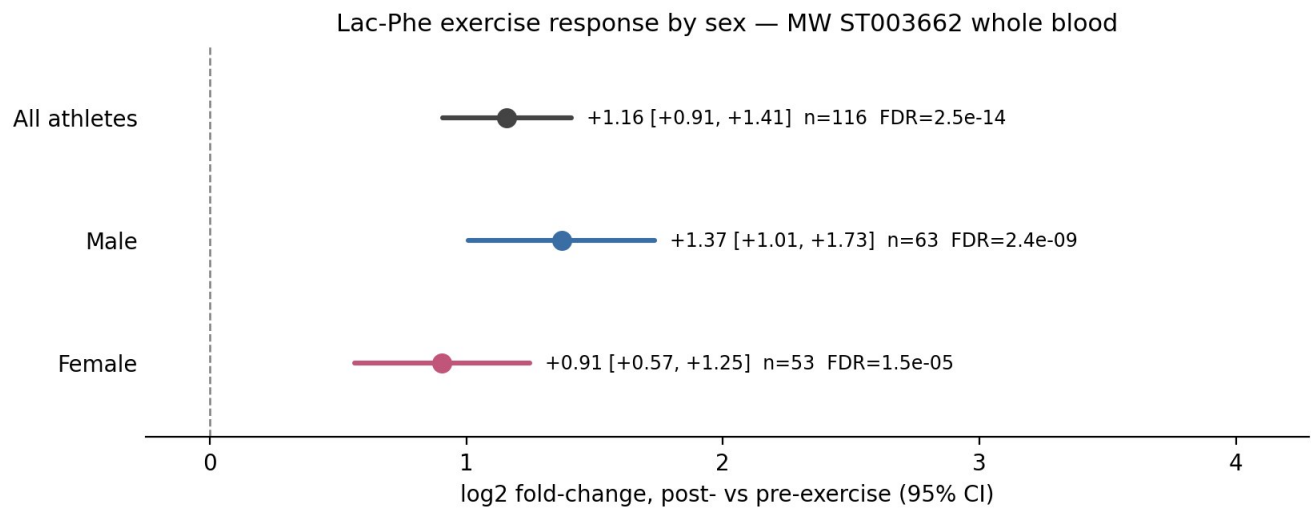


Figure 8. Whole-metabolome volcano per stratum, with Lac-Phe circled.



CIs recovered from the stored paired-t effect, p-value and pair count. Sex strata overlap the 'All' stratum and are not independent replications.

Figure 9. Lac-Phe effect and 95% CI per stratum. The sex strata are nested inside the all-athlete stratum and their intervals overlap.

Rat MoTrPAC pass1b-06: Lac-Phe PRECURSORS only — Lac-Phe itself is absent from the rat feature space

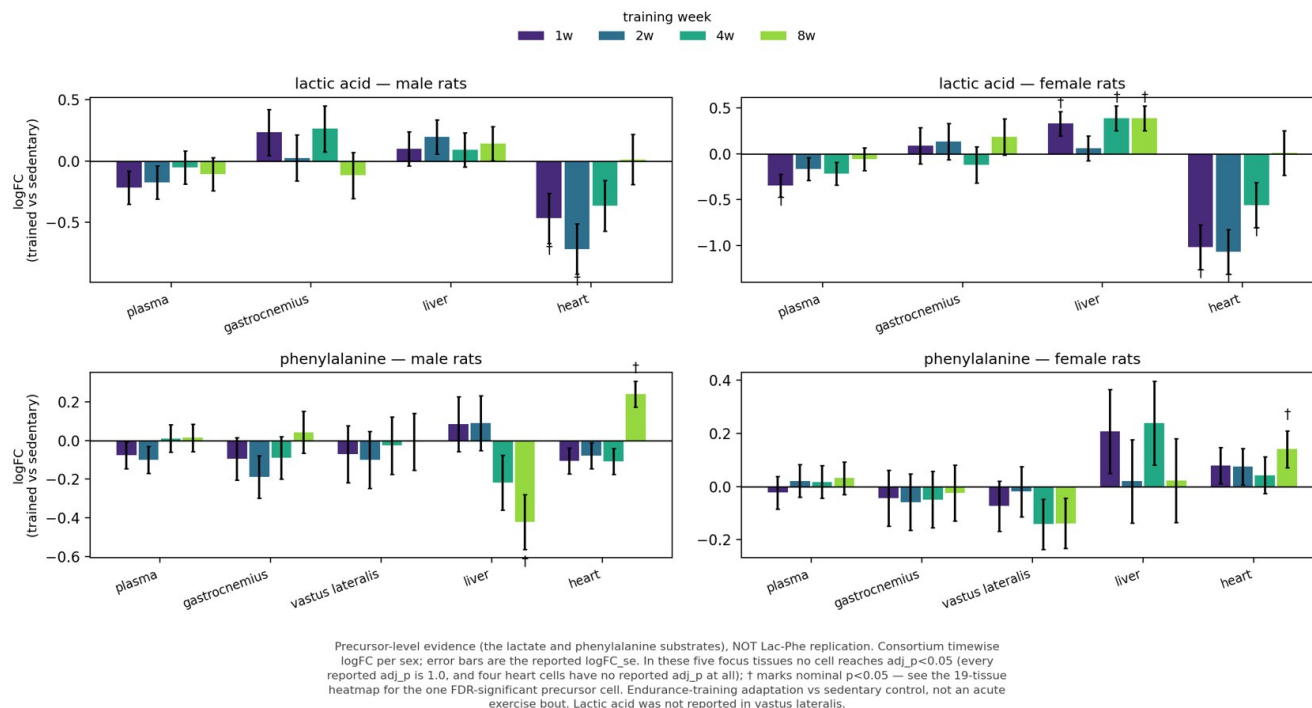


Figure 10. Rat precursor logFC in the five focus tissues, sexes kept separate.

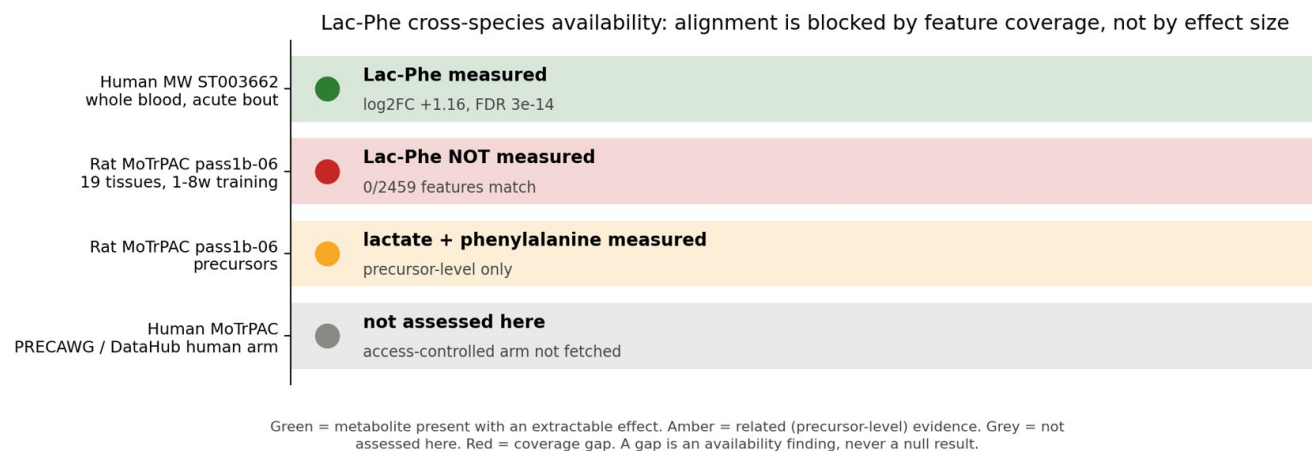


Figure 11. Lac-Phe availability by source. Alignment is blocked by feature coverage, not effect size.

Human MoTrPAC arm

The human MoTrPAC arm was not fetched for this report. The DataHub human arm is access-controlled, and this workbench does not fabricate records for embargoed data. Whether MoTrPAC's human plasma targeted and untargeted panels carry Lac-Phe is an open availability question and is recorded as a mirage risk, not as a negative result. The public human-precovid-sed-adu acute plasma DA tables already cached under data/live/volcano_motrpac_human_plasma_endur_post.csv also contain no lactoyl-conjugate feature.

Human-review escalations

#	item	why a human must decide
1	Assay identity of Lactoyl Phenylalanine in ST003662	RefMet name match only. Level-1 confirmation (RT + MS/MS against an authentic stan...
2	Sex-difference claim	Male vs female point estimates differ but strata are nested with overlapping CIs; ...
3	Exercise protocol	ST003662 Collectionpoint:Before/After does not encode modality, intensity, duratio...
4	Rat panel gap	Confirm against MoTrPAC panel documentation whether Lac-Phe was targeted and lost ...
5	Any pooling or meta-analysis	Blocked. No shared analyte, different estimands, different matrices.
6	Whole blood vs plasma	ST003662 is whole blood; MoTrPAC is plasma. Lac-Phe partitioning between erythrocy...
7	Data deposition for the published human exercise studies	0 of 19 subject-confirmed human exercise records name a repository accession in th...
8	Literature coverage of pubmed	The index was unreachable from this egress, so its coverage is unknown. A reviewer...
9	Subject identity of the 220 name-absent records	The index matched them on full text or relevance ranking, but the queried name is ...

Missing evidence

- Exercise modality, intensity, duration, and post-exercise sampling delay for ST003662.
- MS/MS-level identity confirmation and calibration provenance for the ST003662 Lac-Phe feature.
- Whether rat pass1b-06 panels ever targeted N-lactoyl conjugates.
- Lac-Phe availability in the access-controlled human MoTrPAC arm.
- Independent human exercise studies measuring Lac-Phe in MW (none found beyond ST003662).
- Deposited data for the 19 subject-confirmed human exercise records found in the literature: no accession appears in their retrieved bibliographic text and full texts were not read.
- Study-level identity across those records: companion papers and overlapping cohorts are unresolved, so the count of distinct human exercise cohorts is an upper bound.
- Full-text methods for MSI-level identity and calibration in the published human exercise studies.
- pubmed literature coverage for this query (index unreachable from this egress).

Reproduction

Live steps 1-5 need network access; steps 6-9 are offline. Every step writes under data/live/, reports_live/ or both, and none touches the offline pilot.

```
# 1. resolve the RefMet identity and find every MW study reporting Lac-Phe (network)
PYTHONPATH=src python3 -m metabotyping_agentic.cli live-search-metabolite-studies \
  --query "N-Lactoyl phenylalanine" \
  --name-variants "Lac-Phe;N-lactoylphenylalanine" \
  --out data/live/metabolite_search_lacphe

# 2. same search for all 22 N-lactoyl conjugates in RefMet, for the class-coverage figure (network)
grep -i "^N-Lactoyl" data/live/refmet_annotations.csv | cut -d, -f1 | while read -r name; do
  slug=$(echo "$name" | tr 'A-Z' 'a-z_' | tr -cd 'a-z0-9_')
  PYTHONPATH=src python3 -m metabotyping_agentic.cli live-search-metabolite-studies \
    --query "$name" --out "data/live/metabolite_search_lacphe_family/$slug"
done

# 3. human pre/post effects + participant-level values + rat pass1b-06 coverage scan (network)
PYTHONPATH=src python3 scripts/volcano_compare.py --metabolite-scan "N-Lactoyl phenylalanine" \
  --name-variants "Lac-Phe;N-lactoylphenylalanine;lactoylphenylalanine" \
  --mw-studies ST003662

# 4. the lactate substrate, for the substrate-coupling figure (network; rat scan skipped)
PYTHONPATH=src python3 scripts/volcano_compare.py --metabolite-scan "Lactic acid" \
  --mw-studies ST003662 --scan-skip-rat
```

```
# 5. the published literature for the same subject (network)
PYTHONPATH=src python3 -m metabotyping_agentic.cli live-search-literature \
  --query "N-Lactoyl phenylalanine" \
  --name-variants "Lac-Phe;N-lactoylphenylalanine;lactoylphenylalanine;N-lactoyl-phenylalanine" \
  --out data/live/literature/lacphe --reports-out reports_live/literature

# 6. figure suite + one backing CSV per figure (offline)
python3 scripts/render_lacphe_figures.py

# 7. this report (offline)
python3 scripts/render_lacphe_report.py

# 8. PDF render of this report with the figure plates embedded (offline)
python3 scripts/render_markdown_pdf.py \
  reports_live/lacphe/lacphe_report.md reports_live/lacphe/lacphe_report.pdf

# 9. self-contained HTML render of this report, figures inlined (offline)
python3 scripts/render_markdown_html.py \
  reports_live/lacphe/lacphe_report.md reports_live/lacphe/lacphe_report.html
```

data/live/volcano_motrpac_human_plasma_endur_post.csv is reused from the earlier MoTrPAC human plasma acute fetch rather than re-downloaded; its endpoints and signed-URL requests are recorded in data/live/volcano_provenance.json.

Live fetch provenance:

- data/live/metabolite_scan/n_lactoyl_phenylalanine/provenance.json - human ST003662 scan + rat pass1b-06 coverage scan
- data/live/metabolite_scan/lactic_acid/provenance.json - lactate scan (rat scan skipped, recorded)
- data/live/metabolite_search_lacphe/metstat_metabolite_study_provenance.json - RefMet + metstat search
- data/live/metabolite_search_lacphe_family/<conjugate>/metstat_metabolite_study_provenance.json - 22 conjugate searches
- data/live/volcano_provenance.json - MoTrPAC human plasma acute panel fetch
- data/live/literature/lacphe/literature_provenance.json - literature sweep: per-index status, query expression, endpoint URLs, hit and retrieved counts

gate	status
FAIR provenance	pass - accession, analysis id, source URL, license, matrix, contrast, statistic recorded
reproducibility	pass - regenerated from declared queries and deterministic matching rules
critical evidence	pass - exact vs precursor vs missing evidence kept distinct
human review	pass - 9 report-level escalations plus 419 record-level literature escalations; harmonization blocked
mirage detection	pass - rat gap, embargoed human arm, and the unreachable literature index are reported as availability findings, ne...
literature evidence tiering	pass - peer-reviewed, preprint, secondary-synthesis and retracted records kept in separate tiers; name matches with...